



R V COLLEGE OF ENGINEERING
(An autonomous institution affiliated to VTU, Belgaum)
DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

IM233AI – WORK SYSTEMS DESIGN

UNIT IV

Topic ID 9: Lean and Waste

Part B

Q No	Question	Marks	Difficulty Level	BT Level	CO
1	Explain the "Seven Wastes" (Muda) in a production system with specific focus on defective parts and overproduction.	8	M	2	2
2	Define "Muda" and explain the waste of Overproduction . Why is it considered the most serious waste?	8	M	4	2
3	Compare and contrast "Waste of Motion" and "Waste of Transportation." Provide one industrial example for each and evaluate how excessive inventory contributes to both.	8	H	4	2
4.	Describe the elements of a typical structure of Lean Manufacturing System. Support your answer with an illustration.	8	M	2	2
5.	Define Over-processing as a form of waste. Provide an example from a machining context.	8	M	3	2
6	Explain the waste of Waiting (Idle time). What are typical causes in a manufacturing plant?	8	M	2	2

Topic ID 10: Just in Time Production

7	Compare and contrast Push vs. Pull systems of production control. Why is Pull preferred in Lean?	8	H	4	2
8	Explain the SMED (Single Minute Exchange of Die) methodology for setup time reduction.	8	M	3	2
9	Why is "Setup Time Reduction" a prerequisite for implementing a Pull system?	8	H	4	2
10	Describe the function of a "Kanban" card in a dual-card system with an example.	8	M	3	2
11	Explain the Pull System of production control using the Kanban mechanism.	8	M	2	2
12	Analyze the Internal vs. External Setup activities in the context of SMED.	8	M	4	2



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Topic ID 11: Automation

13	What is Total Productive Maintenance (TPM)? Explain the goal of "Zero Breakdowns."	8	M	2	2
14	Explain the concept of Poka-Yoke (Error Proofing) with three industrial manufacturing examples.	8	H	3	2
15	What is Jidoka? Explain how "Stopping the Process" actually improves long-term efficiency.	8	H	4	2
16	Explain the "Andon" system and its role in the Jidoka philosophy.	8	M	3	2
17	What is Total Productive Maintenance (TPM) ? List its primary goals regarding machine availability.	8	M	2	2

Topic ID 12: Worker Involvement

18	Define "Standardized Work" and explain its three essential components according to Groover.	8	M	2	2
19	Discuss the role of "Takt Time" in aligning production rate with customer demand.	8	M	3	1
20	Explain the 5S System of visual management and its impact on workplace productivity.	8	M	3	1
21	Analyze the organizational requirements for a successful Continuous Improvement culture?	8	H	4	2